

Teaching Exam

Subject – Computer Science

**Computational thinking and
programming-2**



Lecture No -2

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Topics to be Covered

Topic

File Handling

Read / write.



Slide 2: Introduction to Files

- A file stores data on disk
- Two main types:
 - Text Files → .txt, .docx Unicode (String)
 - Binary Files -img
- Files can be opened, read, written, and closed

File I/O performed in any programming language

- Open file —
- Read write operation. =
- File closing —

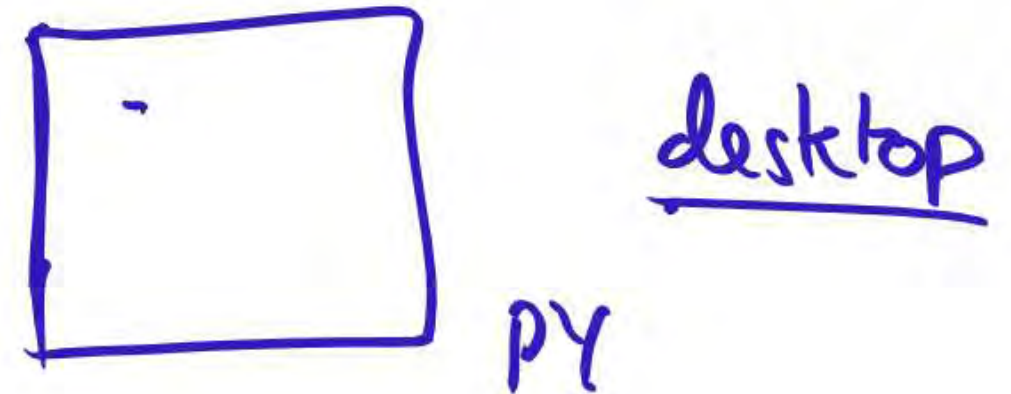
Slide 3: File Paths

✓ Absolute Path: Full path to file ✓ Relative Path: Path relative to current directory

Example:

- Absolute: C:\Users\Documents\file.txt
- Relative: data\file.txt

`f = open("file1.txt", 'r')`



Slide 4: Text Files - Python Basics

✓ Opening a file:

```
f = open("data.txt", "r")
```

✓ Modes:

• r, r+, w, w+, a, a+ ✓ Closing a file:

```
f.close()
```

✓ Using with clause:

```
with open("data.txt", "r") as f:  
    data = f.read()
```

f = open ('data.txt', 'w')
↑ ↑

→ f.read()

→ f.write("Hello")

f = open('data.txt', 'a')

f.write('abc')

Helloabc

Hello

abclo


```
f = open('data.txt', 'w')  
f.write("Hello World")  
f.write("How are you")  
f.close()
```

f.write("Hello World, How are you")

Hello World
How are you.

```
f = open('data.txt', 'w')  
f.write("Hello World")  
f.close()
```

```
f = open('data.txt', 'a')  
f.write("How are you")  
f.close()
```

Hello World
How are you.

How are you.

```
f = open('data.txt', 'a')
```

```
s = f.read()
```

```
print(s)
```

```
f.close()
```

Hello World
How are you.

```
f = open('data.txt', 'a')
```

```
print(f.read(10))
```

```
print(f.read(10))
```

```
f.close()
```

Hello World.

import os.

```
os.remove('data.txt')
```


or+

read. + write.

✓

✓

No truncate.

w_+

✓

✓

Truncate.

a_+

✓

append.

No truncate

Slide 5: Writing/Appending Python

✓ write() ✓ writelines() ✓ Example:

```
with open("data.txt", "w") as f:  
    f.write("Hello World!\n")  
    f.writelines(["Line 2\n", "Line 3\n"])
```

list = ['soni\n',
 'Rohan\n',
 'Aarav\n']

Slide 6: Reading Text Files - Python

✓ read() ✓ readline() ✓ readlines() ✓ Example:

Hello world.
How are you?

```
with open("data.txt", "r") as f:  
    lines = f.readlines()  
    print(lines)
```

f = open(" ", " ")

print(f.read(10)) → Hello world
f.readline() → Hello world
f.readlines()

Slide 7: Seek and Tell - Python

✓ tell() returns cursor position ✓ seek() moves cursor

```
with open("data.txt", "r") as f:  
    print(f.tell())  
    f.seek(0)
```

f.read(10)

f.seek(15)

f.read(5) → 15, 16, 17, 18, 19

print(f.tell()) → 19

Slide 8: Manipulating Text File Data

✓ Read all lines ✓ Modify content ✓ Write back

```
with open("data.txt", "r") as f:  
    lines = f.readlines()  
[lines[0]] = "Updated line\n"  
with open("data.txt", "w") as f:  
    f.writelines(lines)
```

Slide 9: Binary Files - Python

✓ Modes: rb, wb, ab, rb+, wb+, ab+ ✓ Using pickle

- Dump object:

```
import pickle  
with open("data.bin", "wb") as f:  
    pickle.dump(data, f)
```

- Load object:

```
with open("data.bin", "rb") as f:  
    obj = pickle.load(f)
```


Slide 10: CSV Files - Python

✅ Import csv module ✅ Writing:

```
import csv
with open("data.csv", "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerow(["Name", "Age", "Id"])
```

Comma Separated
Files

1) Raju , 32 , 5104.
2
3
4) Ravi , 28 , 6212

✅ Reading:

```
with open("data.csv", "r") as f:
    reader = csv.reader(f)
    for row in reader:
        print(row)
```

Slide 14: Summary Table

File Type	Python Module	Java Class
Text	open()	<u>FileReader, FileWriter</u>
Binary	pickle	<u>ObjectInputStream, ObjectOutputStream</u>
CSV	csv	<u>N/A (use <u>OpenCSV</u> or manual parsing)</u>

1. To open a file c:\scores.txt for reading,
we use _____

- a) infile = open("c:\scores.txt", "r")
- b) infile = open("c:\\scores.txt", "r")
- c) infile = open(file = "c:\scores.txt", "r")
- d) infile = open(file = "c:\\scores.txt", "r")

2. To open a file c:\scores.txt for writing,
we use _____

- a) outfile = open("c:\scores.txt", "w")
- b) outfile = open("c:\\scores.txt", "w")
- c) outfile = open(file = "c:\scores.txt", "w")
- d) outfile = open(file = "c:\\scores.txt", "w")

3. To open a file c:\scores.txt for appending data, we use _____

- a) outfile = open("c:\\scores.txt", "a")
- b) outfile = open("c:\\scores.txt", "rw")
- c) outfile = open(file = "c:\\scores.txt", "w")
- d) outfile = open(file = "c:\\scores.txt", "w")

4. Which of the following statements are true?

- a) When you open a file for reading, if the file does not exist, an error occurs
- b) When you open a file for writing, if the file does not exist, a new file is created
- c) When you open a file for writing, if the file exists, the existing file is overwritten with the new file
- d) All of the mentioned

5. To read two characters from a file object infile, we use

-
- a) `infile.read(2)`
 - b) `infile.read()`
 - c) `infile.readline()`
 - d) `infile.readlines()`

6. To read the entire remaining contents of the file as a string from a file object infile, we use _____

- a) `infile.read(2)`
- b) `infile.read()`
- c) `infile.readline()`
- d) `infile.readlines()`

7. What will be the output of the following Python code?

```
f = None  
for i in range (5):  
    with open("data.txt", "w") as f:  
        if i > 2:  
            break  
print(f.closed)
```

- a) True
- b) False
- c) None
- d) Error

8. To read the next line of the file from a file object infile, we use _____

- a) `infile.read(2)`
- b) `infile.read()`
- c) `infile.readline()`
- d) `infile.readlines()`

9. To read the remaining lines of the file from a file object infile, we use _____

- a) `infile.read(2)`
- b) `infile.read()`
- c) `infile.readline()`
- d) `infile.readlines()`

10. The readlines() method returns _____

- a) str
- b) a list of lines
- c) a list of single characters
- d) a list of integers

1. Which Python function is used to read a single line of text from the user input?

- a) print()
- b) input()
- c) open()
- d) read()

4. Which one of the following is not attributes of file?

- a) closed
- b) softspace
- c) rename
- d) mode

5. What is the use of tell() method in python?
- a) tells you the current position within the file
 - b) tells you the end position within the file
 - c) tells you the file is opened or not
 - d) none of the mentioned

6. What is the current syntax of rename() a file?
- a) rename(current_file_name, new_file_name)
 - b) rename(new_file_name, current_file_name,)
 - c) rename()(current_file_name, new_file_name))
 - d) none of the mentioned

7. What is the current syntax of remove() a file?

- a) remove(file_name)
- b) remove(new_file_name, current_file_name,)
- c) remove(), file_name))
- d) none of the mentioned

9. What is the use of seek() method in files?
- a) sets the file's current position at the offset
 - b) sets the file's previous position at the offset
 - c) sets the file's current position within the file
 - d) none of the mentioned

10. What is the use of truncate() method in file?

- a) truncates the file size
- b) deletes the content of the file
- c) deletes the file size
- d) none of the mentioned



2 mins Summary



TEACHING
WALLAH

Topic

File handling part

THANK YOU